KAIZEN MATHS

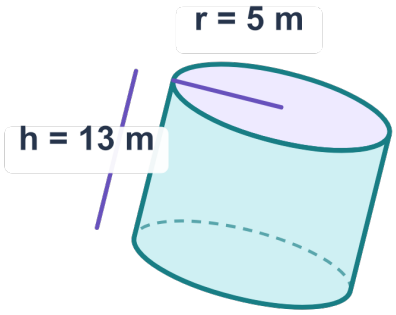
Lesson starter

Cylinders and Cones: Volume and Surface Area

Surface Area

Name _____

SA1



Find total SA

GEO-813A-L2 · SA

Show your working

KAIZEN MATHS

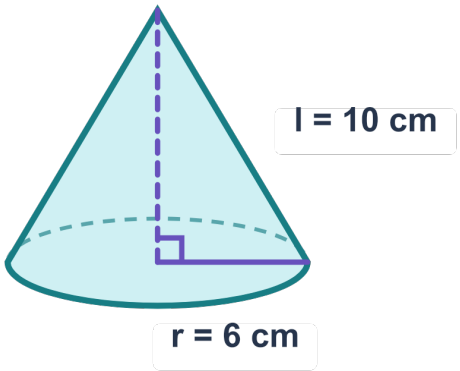
Lesson starter

Cylinders and Cones: Volume and Surface Area

Surface Area

Name _____

SB1



Find total SA

GEO-813A-L2 · SB

Show your working

KAIZEN MATHS

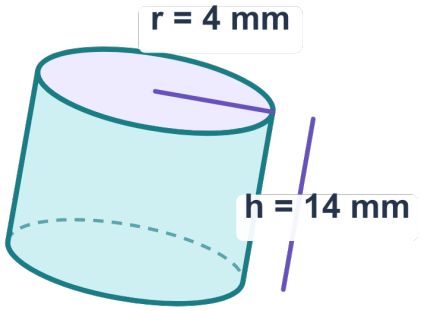
Lesson starter

Cylinders and Cones: Volume and Surface Area

Surface Area

Name _____

SC1



Find total SA

GEO-813A-L2 · SC

Show your working

KAIZEN MATHS

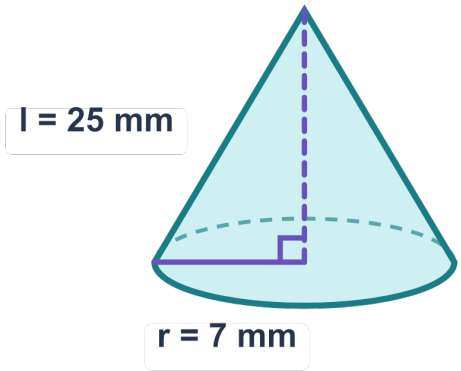
Lesson starter

Cylinders and Cones: Volume and Surface Area

Surface Area

Name _____

SD1



Find total SA

GEO-813A-L2 · SD

Show your working

KAIZEN MATHS

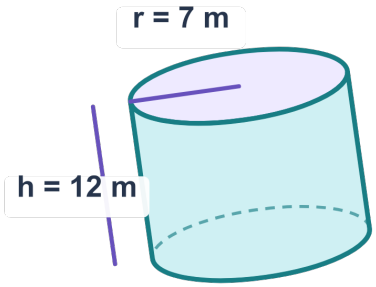
Exit ticket

Cylinders and Cones: Volume and Surface Area

Surface Area

Name _____

EA1



Find total SA

GEO-813A-L2 · EA

Show your working

KAIZEN MATHS

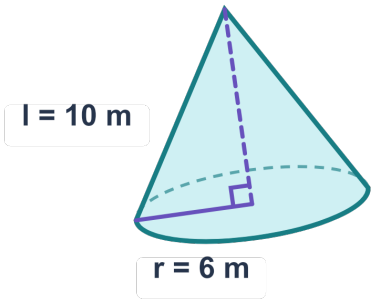
Exit ticket

Cylinders and Cones: Volume and Surface Area

Surface Area

Name _____

EB1



Find total SA

GEO-813A-L2 · EB

Show your working

KAIZEN MATHS

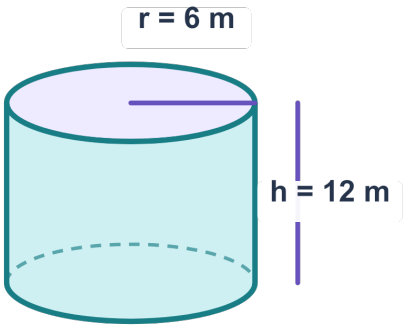
Exit ticket

Cylinders and Cones: Volume and Surface Area

Surface Area

Name _____

EC1



Find total SA

GEO-813A-L2 · EC

Show your working

KAIZEN MATHS

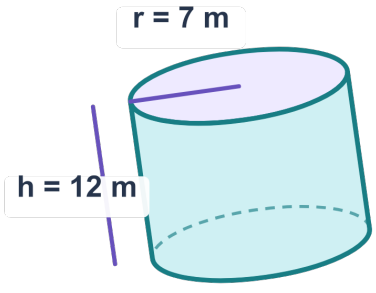
Exit ticket

Cylinders and Cones: Volume and Surface Area

Surface Area

Name _____

EA1



Find total SA

GEO-813A-L2 · EA

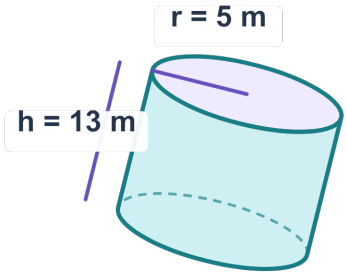
Show your working

Teacher answers & working

Cylinders and Cones: Volume and Surface Area · Surface Area

GEO-813A-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



Find total SA

Answer:

565.49 m^2

Working:

Curved area = $2\pi rh = 2\pi \times 5 \times 13 = 130\pi$ m^2

Two circles = $2\pi r^2 = 2\pi \times 5^2 = 50\pi$ m^2

Total SA = 180π m^2

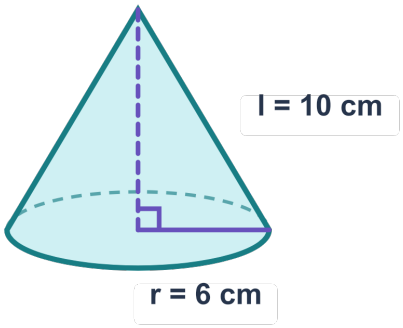
Total SA ≈ 565.49 m^2

Teacher answers & working

Cylinders and Cones: Volume and Surface Area · Surface Area

GEO-813A-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



Find total SA

Answer:

301.59 cm^2

Working:

Curved area = $\pi rl = \pi \times 6 \times 10 = 60\pi$ cm^2

Base area = $\pi r^2 = 36\pi$ cm^2

Total SA = 96π cm^2

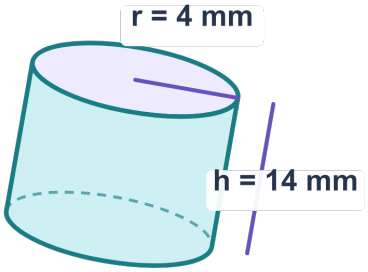
Total SA ≈ 301.59 cm^2

Teacher answers & working

Cylinders and Cones: Volume and Surface Area · Surface Area

GEO-813A-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



Find total SA

Answer:

452.39 mm^2

Working:

Curved area = $2\pi rh = 2\pi \times 4 \times 14 = 112\pi$ mm^2

Two circles = $2\pi r^2 = 2\pi \times 4^2 = 32\pi$ mm^2

Total SA = 144π mm^2

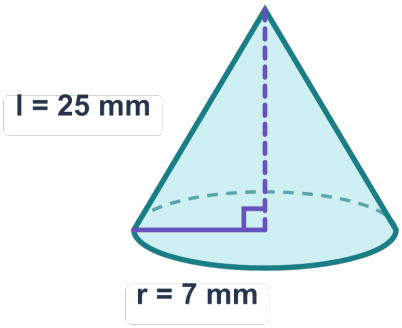
Total SA ≈ 452.39 mm^2

Teacher answers & working

Cylinders and Cones: Volume and Surface Area · Surface Area

GEO-813A-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



Find total SA

Answer:

703.72 mm^2

Working:

Curved area = $\pi rl = \pi \times 7 \times 25 = 175\pi$ mm^2

Base area = $\pi r^2 = 49\pi$ mm^2

Total SA = 224π mm^2

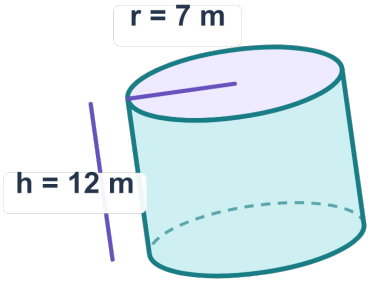
Total SA ≈ 703.72 mm^2

Teacher answers & working

Cylinders and Cones: Volume and Surface Area · Surface Area

GEO-813A-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



Find total SA

Answer:

835.66 m^2

Working:

Curved area = $2\pi rh = 2\pi \times 7 \times 12 = 168\pi$ m^2

Two circles = $2\pi r^2 = 2\pi \times 7^2 = 98\pi$ m^2

Total SA = 266π m^2

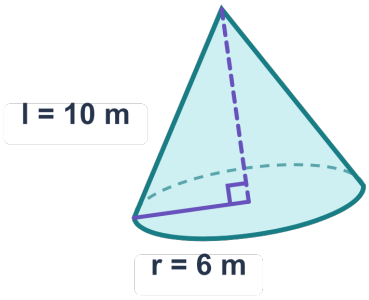
Total SA ≈ 835.66 m^2

Teacher answers & working

Cylinders and Cones: Volume and Surface Area · Surface Area

GEO-813A-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



Find total SA

Answer:

301.59 m^2

Working:

Curved area = $\pi rl = \pi \times 6 \times 10 = 60\pi$ m^2

Base area = $\pi r^2 = 36\pi$ m^2

Total SA = 96π m^2

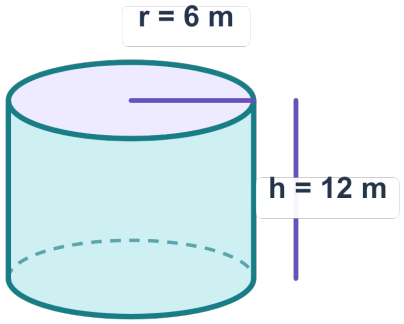
Total SA ≈ 301.59 m^2

Teacher answers & working

Cylinders and Cones: Volume and Surface Area · Surface Area

GEO-813A-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



Find total SA

Answer:

678.58 m^2

Working:

Curved area = $2\pi rh = 2\pi \times 6 \times 12 = 144\pi$ m^2

Two circles = $2\pi r^2 = 2\pi \times 6^2 = 72\pi$ m^2

Total SA = 216π m^2

Total SA ≈ 678.58 m^2